Ilyas Mallah

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Third-year Computer Science student at Utrecht University. I work on performance and correctness across the stack: LLM inference engines (vLLM, llama.cpp), compilers and runtimes (Luau, Wasmtime), GPU kernels, and production backends. Open to internships and contract work, part-time during the year or full-time in the summer.

OPEN SOURCE

Luau (Roblox's scripting language, under the handle `green-real`). 7 changes merged in 2026 into the x64 and AArch64 native code generator, the register allocator and the type checker.

- Integer == in branches: a new IR command with x64 and AArch64 lowering. 3.18 ns to 0.63 ns per comparison (#2571).
- Fixed an x64 register allocator miscompile when restoring a spilled value (#2987).
- Array fast path for `for ... in pairs` on AArch64, matching x64 (#2683).

llama.cpp (CUDA, LLM inference). Traced why decoding with a q4_0 KV cache ran at half speed on RTX 50 series GPUs with CUDA 12.8 to a compiler bug, and worked around it in the kernel: up to 2.1x faster decode at 32k context on an RTX 5090 (#30077, in review).

LLVM (libc++, Clang). Twelve PRs in review (checked 2026-10-27): hardening assertions for `insert`/`emplace` positions (#230490), `resize_and_overwrite` bounds (#230605), `valarray` sizes and the elements a subscript selects (#230606, #236120), iterator ranges in `string` `insert`, `append` and `assign` (#234566), `list::splice` and `erase(end())` (#232538, #232661), negative counts in `views::drop`, `views::take` and `views::counted` (#234068), a missing XMM register save in x86 variadic functions built with stack clash protection (#234871), a Clang warning (#230491), and a Clang Safe Buffers gate and its libc-call subgroups (#232898, #233604). Merged: the callee pop size of x86 tail calls with 16-byte aligned stack arguments (#235417).

Wasmtime (WebAssembly runtime, Cranelift code generator). Added float to 8- and 16-bit integer conversions to the x64 backend, which panicked on them, closing an issue open since 2022 (#14778). Made `fpromote` and `fdemote` with f16 or f128 operands fail cleanly, where aarch64 compiled them to wrong values and x64 and s390x panicked (#14883). Both merged in October 2026. Credited reporter of security advisory GHSA-mghf-wjfp-p5xm (CVE-2026-115893), published in October 2026 and fixed in Wasmtime 36.0.19, 48.0.7 and 49.0.4.

wabt (WebAssembly tools). Fixed wasm2c on 32-bit hosts, where a 4 GiB memory size wrapped to 0 and freed memory still in use (#2874). Fixed four wabt tools that exited 0 when they could not write their output file, so they now exit with an error (#2889). Both merged in October 2026.

mruby (embedded Ruby). Fixed `pack` and `unpack` string lengths that wrapped negative at 2 GiB and read outside the string (#7692). Fixed `Float("0")` raising an error, a regression since 4.1.0-rc (#7701). Fixed `Float#to_r` and `Rational(float)` raising `ZeroDivisionError` for very small floats (#7703). All three merged in October 2026.

wgpu (Rust graphics, naga shader translator). Fixed naga's fuzz build, broken on OSS-Fuzz since March 2026. Merged in October 2026 (#10637).

tt-metal (Tenstorrent AI accelerators). Found a float32 accuracy bug in `ttnn.logit` with a simulator sweep of 54 ops, up to 8,767 ULP off near 0.5. My fix brings it to 3 ULP (issue #59930).

vLLM (LLM serving, four PRs in review). On Hopper, picked the CUTLASS block-FP8 kernel by batch size: 17 to 52% more serving throughput on an H100 NVL with Qwen3-8B FP8 (#60464). Stopped block-FP8 models from crashing at startup when nvcc is older than 12.9 (#60438). Removed a residual copy per layer that a compiler pass left in block-FP8 models (#60481). Added a context window for EAGLE-3 speculative decoding drafters: decode at 32k tokens goes from 16 to 34 tok/s on an A40 with Llama-3.1-8B (#63245).

PROJECTS

Ulana, an optimizing WebAssembly to Luau compiler, written in Luau (about 39,000 lines of source). Ten real programs, among them SQLite, QuickJS, jq and zstd, run as generated Luau and match wasmtime's output.

Ganglion, a Rust and CUDA simulator for brain wiring diagrams (connectomes). It runs a fruit fly central nervous system (165,122 neurons, 24.5 M connections) faster than real time on one RTX 3070 Ti. One simulated second takes 0.37 s. Its reference mode matches the Python reference simulator bit for bit.

Lute FFI, a C foreign function interface for the Lute runtime, from specification to a five-milestone implementation with full Luau types. Not yet public: the proposal to the Lute maintainers is in preparation.

EXPERIENCE

Rolantir, analytics platform for the Roblox games market. Co-founder since June 2026. Its catalog holds over 1 million Roblox games in a 24.5 GB production database (October 2026). I built almost all of the backend (TypeScript, PostgreSQL and TimescaleDB). I moved the live 13.9 GB SQLite database to TimescaleDB with the full history kept. Hourly chart queries went from 400 to 900 ms to 3 to 8 ms. Fixed a recurring production deadlock.

Dig It, Roblox game. Co-founder, October 2024 to March 2025. Programming, design and live operations for a game that reached over 73 million visits. At its peak it had 27,400 players online at once, 500,000 daily and 5.3 million monthly active users. Led the negotiation when the game was sold.

EDUCATION

Utrecht University, BSc Computer Science. September 2024 to early 2028 (expected). Average grade 7.71 out of 10. Minor: Entrepreneurship.

SKILLS

- **Languages:** Luau, C++, Rust, CUDA, Python, TypeScript, C#
- **Areas:** compilers and JITs (IR, register allocation, x64 and AArch64 code generation), GPU performance, WebAssembly, databases (PostgreSQL, TimescaleDB)
- **Tools:** Nsight Compute, perf, Git, CMake, Docker
- **Spoken:** Dutch, English